

Issues and possible solutions

The current challenges stem from the fact that obtaining positions requires the double integration of acceleration, a process that inherently promotes the accumulation of errors. An additional source of error is the noise detected by the IMU. As demonstrated in the following video

<https://www.youtube.com/watch?v=C7JQ7Rpwn2k> 23:21 , simple filtering—for example, using a low-pass filter—is insufficient to solve this problem.

More sophisticated solutions are necessary, such as implementing a Kalman filter or integrating data from a larger number of sensors (e.g., a magnetometer), as proposed in the report: <https://www.cl.cam.ac.uk/techreports/UCAM-CL-TR-696.pdf> This report focuses specifically on the errors in navigation based on accelerometers and gyroscopes and may prove valuable in subsequent stages.

If we decided to use a Kalman filter for example, it would be needed to create specific model of movement of IMU on the hexapod which seems to be quite hard